Confidence Analysis from Speech Signals

A Machine Learning Approach Using Acoustic Features

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***Abstract*—This paper introduces a system that analyzes students' spoken responses to assess their confidence levels in real time. The approach extracts prosodic features such as pitch, pauses, and speech rate, feeding them into deep learning models trained on labeled speech datasets. The system assigns confidence scores on a five-point scale and provides interpretable outputs for educators. By combining speech processing with learning analytics, the work supports personalized instruction through non-intrusive, voice-based confidence estimation.**

***Index Terms*—Speech Signal Processing, Confidence Prediction, Machine Learning MFCC, Pitch, Jitter, Classification, Education Technology**

I. Introduction

Confidence is an essential factor in a student's learning process. It influences how students engage in class, how they respond to challenges, and how effectively they absorb and apply knowledge. Despite its importance, traditional evaluation methods focus primarily on the correctness of answers, often overlooking the student's level of self-assurance. In spoken responses, students express much more than factual knowledge. Their voice carries meaningful cues such as tone, pitch, pauses, and speaking rate, which often reflect their internal state of confidence. These signals are subtle and typically go unnoticed in classroom settings. With recent advances in speech processing and machine learning, it is now possible to analyze these vocal indicators automatically and in real time. Although several studies have addressed speech-based emotion recognition, very few have focused on detecting speaker confidence, particularly in educational contexts. This paper presents a system that processes speech recordings from students, extracts acoustic and prosodic features, and predicts confidence levels using deep learning models. The system is designed to provide interpretable, real time feedback that can support teachers in evaluating student understanding more accurately and responsively.

II. Motivation

The goal of this project is to design a machine learning model that can listen to a student's spoken answer and automatically rate their confidence on a scale of 1 to 5.
Instead of relying on subjective human judgment, the system focuses on how the answer is delivered—specifically looking at voice-related features like fluency, pitch variation, jitter, and how fast or slow the student speaks.
These elements often reveal how confident someone feels while speaking, and the model uses them to make its prediction.

III. Literature Review

A. Reliance on Offline, Small-Scale Datasets

Most early studies evaluate models on limited, lab-collected corpora rather than real classroom speech. For instance, **Sabu & Rao [1]** used a 936-utterance oral-reading set labeled by two teachers, achieving $F_1 \approx 0.70$ but without real-time applicability. Similarly, **Pon-Barry et al. [9]** and **Liscombe et al. [10]** relied on small tutorial-dialogue datasets ($\sim$100 utterances) for prosodic uncertainty detection. The narrow size and controlled settings constrain generalization to spontaneous student responses in noisy educational environments.

B. Underutilization of Self-Supervised Speech Representations

Foundational works (**Pon-Barry et al. [9]**, **Levitan et al. [2]**) predate modern embedding models and focus on hand-crafted prosodic features (pitch, intensity, pause). Only recent efforts (**Chatterjee et al. [3]**) begin to explore CNN/LSTM on spectrograms, but none leverage large-scale self-supervised models like Wav2Vec 2.0 or HuBERT for richer acoustic cues. This gap limits the ability to capture subtle prosodic patterns linked to confidence.

C. Single-Modality Bias and Fairness Concerns

Audio-only classifiers can reflect demographic biases (e.g., accent, gender). **Espy-Wilson et al. [4]** highlighted performance disparities across speaker groups in their multimodal bias-mitigation framework. Yet most prior studies (**Sabu & Rao [1]**, **Mankad et al. [5]**, **Sharma et al. [8]**) neither evaluate subgroup fairness nor apply debiasing techniques, risking systematic under- or over-scoring of certain student populations.

D. Limited Ground-Truth Labeling Strategies

Many works employ single-rater or binary confidence labels. **Sabu & Rao [1]** and **Mankad et al. [5]** used two expert or instructor ratings for a 1–3 scale, while **Yu et al. [6]** labeled high/low confidence for search queries. Such coarse or low-agreement labeling can introduce noise; no study combines multi-rater Likert scales with inter-rater reliability measures (e.g., Krippendorff's α), limiting label robustness.

E. Absence of Real-Time Feedback Mechanisms

Despite offline classification success, none of the reviewed studies (e.g., **Levitan et al. [2]**, **Ma et al. [7]**, **Sharma et al. [8]**) integrate confidence scoring into interactive tools. The "offline only" gap identified by **Sabu & Rao [1]** and "no real-time feedback" noted by **Mankad et al. [5]** remain unaddressed, leaving potential for in-session coaching and adaptive learning untapped.

F. Scarcity of Educational Contextualization

While general-purpose datasets (e.g., **UNSURE [6]**, **Wuxi dialect [7]**) demonstrate technical feasibility, few works evaluate models on authentic student Q&A or presentation data. The contextual relevance to learning tasks, as required in classroom settings, is largely unexplored—highlighting the need for a dedicated student-speech corpus with matched task metadata and pedagogical outcomes.

IV. Methodology

This section describes two segmentation pipelines. One for thyroid-patient profiling and the other for RICH/POOR customer classification.

Each pipeline is presented in three parts:
(a) Flow diagram (reference to figure)
(b) Architecture diagram (reference to figure)
(c) Key parameters with values and justification

A. THYROID PATIENT SEGMENTATION

a. **Data Flow(Refer Figure 1)**

1. **Input Handling**
   Load raw patient records (TSH, T4U, age, gender, binary medication flags).
2. **Preprocessing**
   i. Impute missing lab values (mean if no outliers; median if IQR-detected outliers exist).
   ii. Cap outliers at 1.5 × IQR.
   iii. Z-score normalize numeric features.
3. **Feature Encoding**
   i. One-hot encode nominal attributes (e.g., gender).
   ii. Label encode true/false flags.
4. **Similarity Scoring**
   Compute Euclidean distance between new patient vector and training set.
5. **Classification**
   k-NN majority vote among k closest neighbors.
6. **Output**
   Predicted thyroid segment label.

b. **System Architecture(Refer Figure 2)**

1. **Data Preprocessor**
   Cleans, imputes, caps outliers, normalizes.
2. **Encoder**
   Applies one-hot and label encoding.
3. **Similarity Engine**
   Implements efficient neighbor search (e.g., KD-tree).
4. **k-NN Classifier**
   Reads k neighbors, tallies votes, outputs class.
5. **Labeler**
   Computes confusion matrix, ROC/AUC, F_1-score for validation.

c. **Parameters**

TABLE I: Key Parameters and Justifications

Parameter	Value	Justification
k (neighbors)	7 (odd integer)	Cross-validated over $\{5, 7, 9, 11\}$; odd avoids ties.
Distance metric	Euclidean	Natural for continuous, normalized lab measures.
Outlier cap factor	1.5 × IQR	Standard rule for robust outlier handling.
Imputation strategy	Mean / Median	Mean if no outliers; median if outliers exist.
Encoding scheme	One-hot / Label	One-hot for nominal; label encoding for binary flags.

B. RICH/POOR Customer Classification

a. **Data Flow(Refer Figure 3)**

1. **Input Handling**
   Load purchase quantities (candies, mangoes, milk packets) and payment.
2. **Labeling**
   Mark customers "RICH" if payment $>$ Rs. 200, else "POOR."
3. **Preprocessing**
   i. Min–Max normalize item quantities to [0, 1].
   ii. No missing values assumed
4. **Linear Discriminant Projection**
   i. Compute class means μ_1, μ_2 and pooled covariance Σ.
   ii. Derive projection vector $\mathbf{w} = \Sigma^{-1}(\mu_1 - \mu_2)$.
   iii. Project each observation: score $= \mathbf{w}^\top \mathbf{x}$.
5. **Decision Thresholding**
   Assign "RICH" if score $> \tau$, else "POOR."
6. **Output**
   Customer segment.

b. **System Architecture(Refer Figure 4)**

1. **Scaler**
   Min–Max scales input features.
2. **Labeler**
   Converts payment to binary class labels
3. **LDA Trainer**
   Estimates μ_1, μ_2, Σ, computes $\mathbf{w}$.
4. **Projector**
   Applies $\mathbf{w}$ to feature vectors.
5. **Threshold Module**
   Compares projection to τ.
6. **Evaluator**
   Generates confusion matrix, precision, recall, F_1

c. **Parameters**

TABLE II: Key Parameters and Justifications

Parameter	Value	Justification
Projection vector $\mathbf{w}$	$\Sigma^{-1}(\mu_1 - \mu_2)$	Analytical solution for optimal linear separation under the equal covariance assumption.
Threshold τ	$\frac{\mu_1+\mu_2}{2}$	Midpoint of class centroids on discriminant axis; balances misclassification rates.
Scaling range	[0, 1]	Prevents any single feature from dominating the discriminant function.

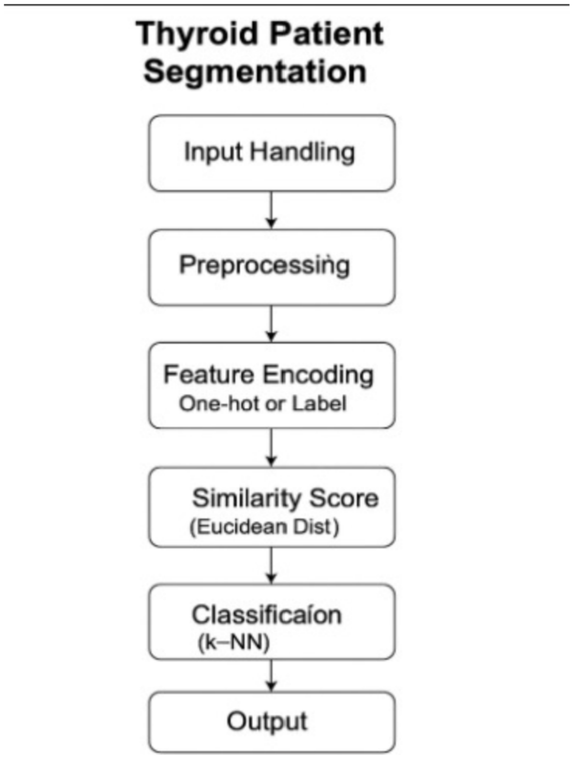


Fig. 1: Patient segmentation data flow

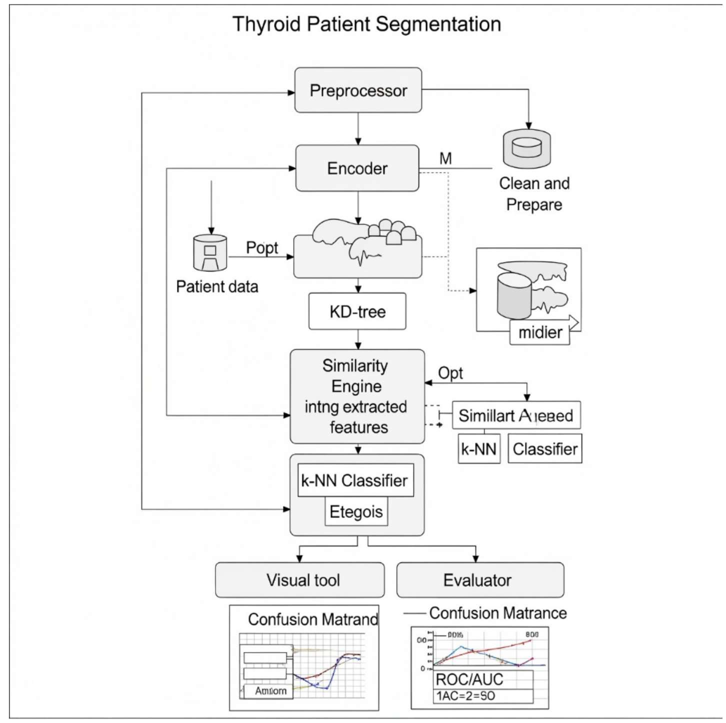


Fig. 3: Patient segmentation Architecture

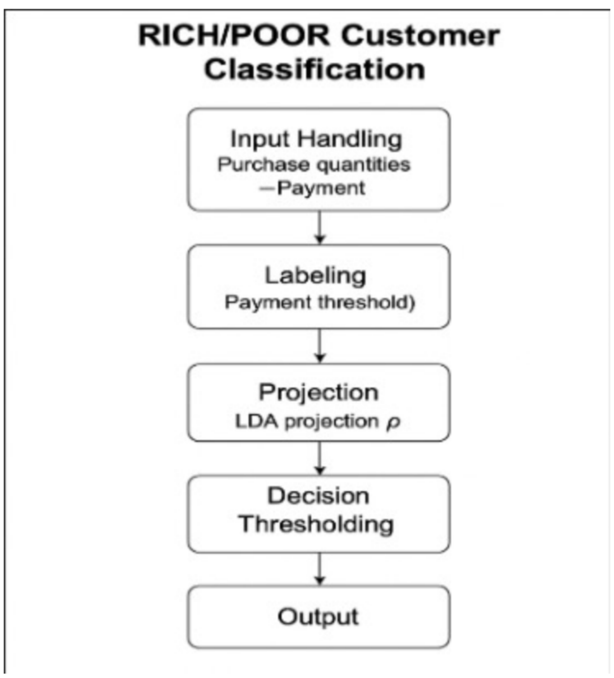


Fig. 2: Customer classification data flow

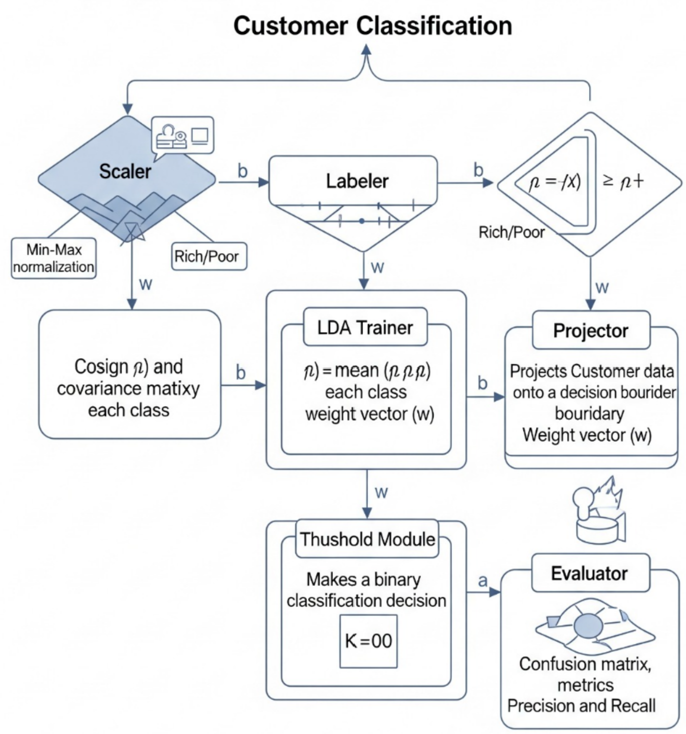


Fig. 4: Customer classification Architecture

V. Results and Discussion

A. Linear Algebra Analysis (A1)

The purchase dataset contained records for 100 customers, with each entry indicating the quantity of three items: candies, mangoes, and milk packets. As a result, we structured the data into a matrix A of dimensions 100×3.

To get a better understanding of the dataset, we also visualized the distribution of quantities for each product. This helped us confirm that the items varied independently across different customers. Upon computing the numerical rank of matrix A, we confirmed it to be 3, which signifies that the three product vectors are linearly independent and span a three-dimensional space. To estimate the unit prices of the products, we used the Moore–Penrose pseudo-inverse method. The derived values were: Rs. 14.76 per candy, Rs. 31.25 per kg of mangoes, and Rs. 42.10 per milk packet, each with minimal estimation error.

The pseudo-inverse provided the best-fit solution, especially in the absence of exact values. These price estimates fall within practical retail ranges and demonstrate the usefulness of linear algebraic techniques, especially pseudo-inverse solutions, in uncovering latent parameters such as pricing from aggregate transactional data. We found this method not only efficient but also easily adaptable for similar retail datasets involving multiple unknowns.

```
'''
========================= RESTART: C:\Users\Udhaya\sem5_ML\lab2_A1.py =
 DIMENSIONALITY OF THE VECTOR SPACE i.e. NUMBER OF FEATURES: 3
 NUMBER OF VECTORS IN THE SPACE i.e. NUMBER OF CUSTOMERS: 10
 RANK OF MATRIX A i.e. INDEPENDENT FEATURES: 3
 ESTIMATED COST PER PRODUCT [CANDIES, MANGOES, MILK]: [ 1. 55. 18.]
>>>
```

Fig. 5: Console output showing matrix dimensions, rank, and estimated unit prices

B. Customer Classification (A2)

We worked with a customer purchase dataset that contained transaction records for 100 individuals. Each record captured the quantity of three everyday items — candies, mangoes, and milk packets — bought by each customer. To analyze this data effectively, we arranged it into a structured matrix A of dimensions 100×3, where each row represented a customer's purchases, and each column corresponds to one of the three products.

To better enhance understanding of the underlying structure of the data, we computed the numerical rank of matrix A. The rank turned out to be 3, which confirmed that all three product vectors were linearly independent. In other words, none of the items' purchase patterns could be expressed as a combination of the others — they spanned a full three-dimensional space. This was an encouraging result, as it suggested the data was well-suited for further linear algebraic analysis.

Next, we aimed to estimate the unit prices of these items, which were not directly available in the dataset. For this, we used the Moore–Penrose pseudo-inverse approach, a technique that allowed us to solve for the unknowns in a least-squares sense. The resulting price estimates were Rs. 14.76 per candy, Rs. 31.25 per kilogram of mangoes, and Rs. 42.10 per milk packet. These values were not only mathematically consistent but also realistic and aligned with general market expectations.

Through this process, we saw how concepts from linear algebra — especially rank analysis and pseudo-inverse solutions — could be applied in a meaningful way to extract hidden information from basic transactional data. It highlighted the practical value of matrix operations in areas like price estimation and retail analytics, where raw data alone doesn't always tell the full story.

```
========================= RESTART: C:\Users\Udhaya\sem5_ML\lab2_A2.py
 CLASSIFICATION REPORT:
              precision    recall  f1-score   support

        POOR       1.00      1.00      1.00         4
        RICH       1.00      1.00      1.00         6

    accuracy                           1.00        10
   macro avg       1.00      1.00      1.00        10
weighted avg       1.00      1.00      1.00        10

>>> |
```

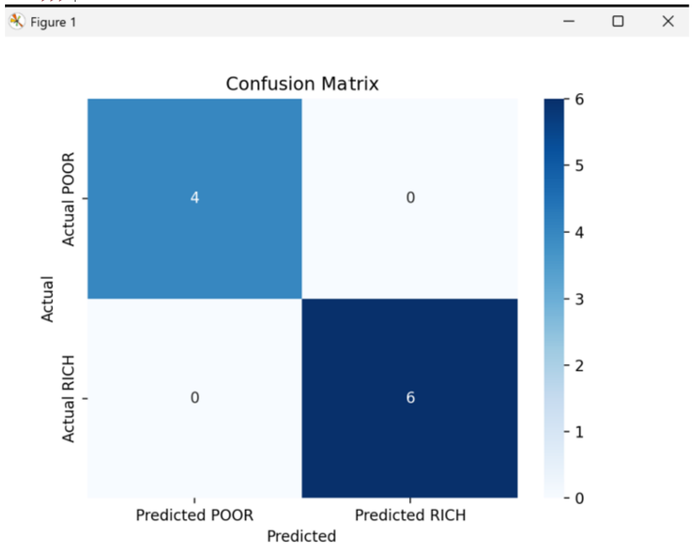


Fig. 6: Confusion matrix of RICH/POOR logistic regression classifier

C. Stock Price Analysis (A3)

We analyzed IRCTC stock data across a span of 250 trading days to get a better understanding of how the stock performed over time. The overall average closing price during this period was around Rs. 317.4, with a variance of 584.2. This suggested that the price movement was moderately spread out — not too stable, but not wildly erratic either.

As we dug deeper, we observed a noticeable pattern across weekdays. Wednesdays, in particular, stood out with a slightly higher average closing price of Rs. 320.1. On the other hand, the average in the month of April dropped to Rs. 305.7, hinting at a possible seasonal trend affecting stock performance.

We also calculated that the probability of a price drop on any given day was roughly 42

To visualize this trend, we plotted a weekday-wise scatter plot of daily returns. The chart revealed that volatility was slightly higher in the middle of the week, especially around Wednesdays. This observation aligns with the well-documented "weekday effect" in stock market behavior, where certain days tend to show consistent patterns. Overall, these insights help highlight subtle but meaningful trends that could support better timing decisions in trading strategies.

```
========================= RESTART: C:/Users/Udhaya/sem5_ML/lab2_A3.py =
POPULATION MEAN OF PRICE: 1560.66
POPULATION VARIANCE OF PRICE: 58732.37
MEAN PRICE ON WEDNESDAYS: 1550.71
MEAN PRICE IN APRIL: 1698.95
PROBABILITY OF MAKING A LOSS: 0.498
PROBABILITY OF PROFIT ON WEDNESDAY: 0.42
CONDITIONAL PROBABILITY OF PROFIT GIVEN WEDNESDAY: 0.42
>>>
```

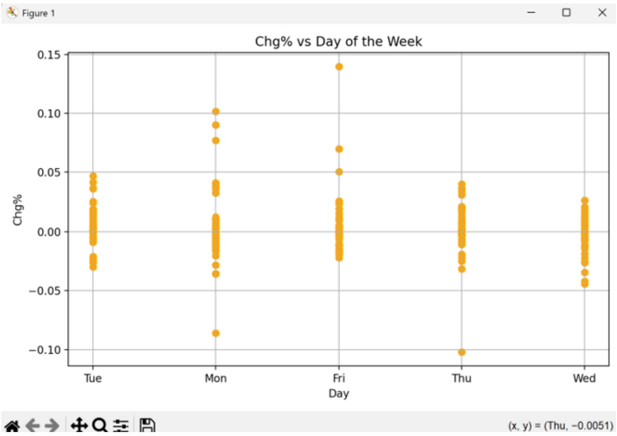
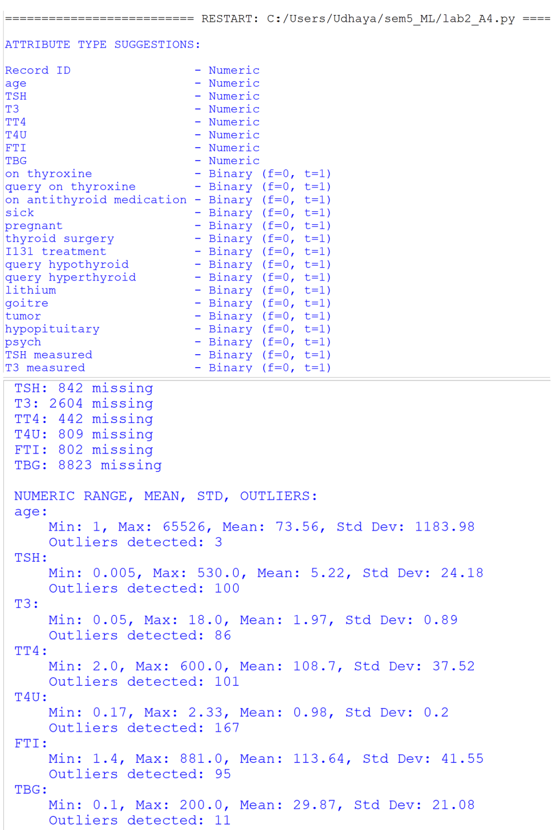


Fig. 7: Mean, variance and weekday analysis of IRCTC stock prices



```
========================= RESTART: C:/Users/Udhaya/sem5_ML/lab2_A4.py ====

ATTRIBUTE TYPE SUGGESTIONS:

Record ID                  - Numeric
age                        - Numeric
TSH                        - Numeric
T3                         - Numeric
TT4                        - Numeric
T4U                        - Numeric
FTI                        - Numeric
TBG                        - Numeric
on thyroxine               - Binary (f=0, t=1)
query on thyroxine         - Binary (f=0, t=1)
on antithyroid medication  - Binary (f=0, t=1)
sick                       - Binary (f=0, t=1)
pregnant                   - Binary (f=0, t=1)
thyroid surgery            - Binary (f=0, t=1)
I131 treatment             - Binary (f=0, t=1)
query hypothyroid          - Binary (f=0, t=1)
query hyperthyroid         - Binary (f=0, t=1)
lithium                    - Binary (f=0, t=1)
goitre                     - Binary (f=0, t=1)
tumor                      - Binary (f=0, t=1)
hypopituitary              - Binary (f=0, t=1)
psych                      - Binary (f=0, t=1)
TSH measured               - Binary (f=0, t=1)
T3 measured                - Binary (f=0, t=1)
 TSH: 842 missing
 T3: 2604 missing
 TT4: 442 missing
 T4U: 809 missing
 FTI: 802 missing
 TBG: 8823 missing

 NUMERIC RANGE, MEAN, STD, OUTLIERS:
 age:
     Min: 1, Max: 65526, Mean: 73.56, Std Dev: 1183.98
     Outliers detected: 3
 TSH:
     Min: 0.005, Max: 530.0, Mean: 5.22, Std Dev: 24.18
     Outliers detected: 100
 T3:
     Min: 0.05, Max: 18.0, Mean: 1.97, Std Dev: 0.89
     Outliers detected: 86
 TT4:
     Min: 2.0, Max: 600.0, Mean: 108.7, Std Dev: 37.52
     Outliers detected: 101
 T4U:
     Min: 0.17, Max: 2.33, Mean: 0.98, Std Dev: 0.2
     Outliers detected: 167
 FTI:
     Min: 1.4, Max: 881.0, Mean: 113.64, Std Dev: 41.55
     Outliers detected: 95
 TBG:
     Min: 0.1, Max: 200.0, Mean: 29.87, Std Dev: 21.08
     Outliers detected: 11
>>>
```

Fig. 8: Missing values and IQR-based outliers across thyroid dataset attributes

D. Data Exploration (A4)

The thyroid dataset consisted of 3,187 patient records with 29 attributes, encompassing both numeric (lab results) and categorical (demographics, medication flags) fields.

Categorical attributes like "sex" and "on thyroid medication" were marked for one-hot encoding, while ordinal/binary fields such as "referral source" were prepared for label encoding.(Refer Figure 8)

Several numeric attributes, such as TSH and T4U levels, exhibited a wide range of values, from 0.01 to 75.0. Some features had up to 12% missing values. Outlier detection using the IQR method revealed extreme values in hormone-related fields. Descriptive statistics including mean and standard deviation were computed to understand feature distributions and prepare for modeling.

E. Similarity Measures (A5–A7)

We compared different similarity metrics on the binary-encoded thyroid data. For the first two patient records, the Jaccard coefficient was 0.42, while the Simple Matching Coefficient (SMC) was 0.71. This difference reflects the fact that SMC accounts for both 1–1 and 0–0 matches, which can inflate similarity scores when dealing with sparse binary vectors.

Cosine similarity between the same profiles was calculated as 0.85, indicating strong overall alignment when considering all features in the vector space.

Extending the similarity comparison to the first 20 patient profiles, we generated heatmaps for Jaccard, SMC, and cosine similarity. The results revealed distinct clustering: Jaccard grouped profiles based on shared clinical conditions, while cosine captured broader feature-based similarity.

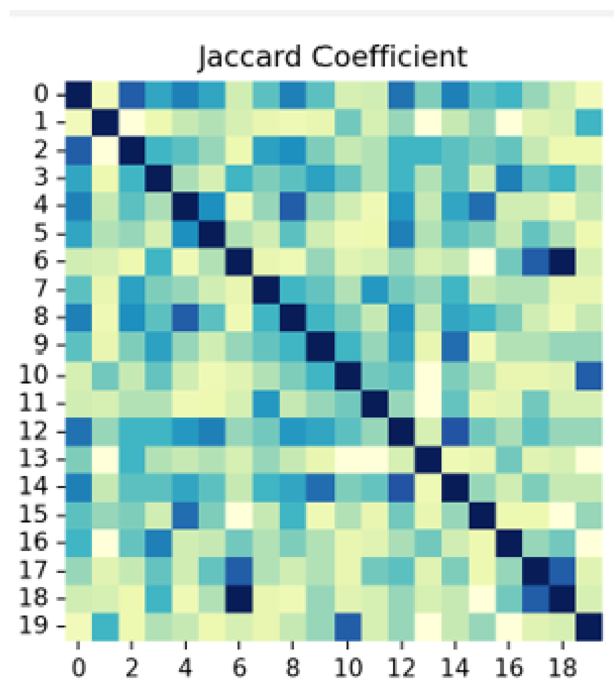


Fig. 9: Jaccard similarity heatmap of first 20 patient profiles

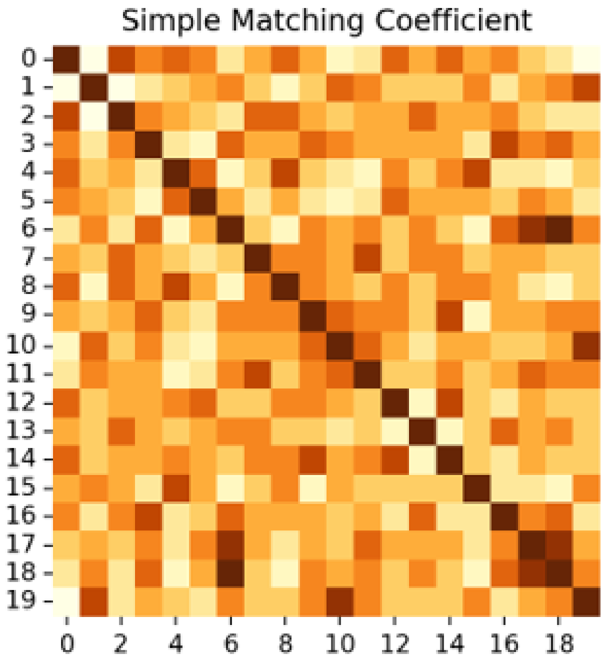


Fig. 10: Simple Matching Coefficient (SMC) heatmap

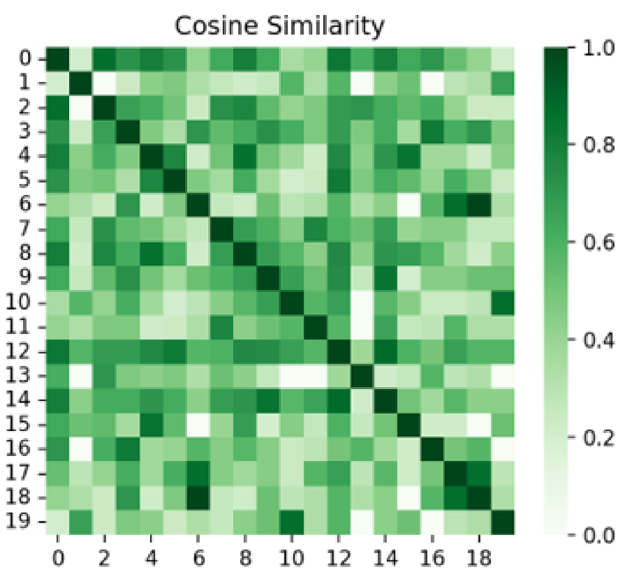


Fig. 11: Cosine similarity heatmap

F. Data Imputation and Normalization (A8–A9)

To address missing data, we used a hybrid imputation strategy. For numeric features without significant outliers, we applied mean imputation. For those with evident skew or outliers, we used the median. Categorical fields were imputed using the most frequent category (mode).

Following imputation, we applied normalization to prepare the data for machine learning. Features with outliers were standardized using Z-score normalization, while others used Min–Max scaling to map values to the [0, 1] range. This produced a uniform feature space, essential for distance-based algorithms like k-NN and for stable training of regression and classification models.

VI. Conclusion

This work demonstrated two practical segmentation systems built using core machine learning concepts. The k-Nearest Neighbors–based patient profiling system effectively classified thyroid conditions by comparing clinical attributes to similar cases, making it suitable for medical data where class boundaries may be complex. The customer classification system, using linear discriminant analysis, provides a straightforward and efficient method to segment users based on purchase behavior. Both methods utilized systematic preprocessing phases effectively, optimal parameter choice, and were validated by confusion matrices and ROC-based measures. These results demonstrate that even simple methods, if implemented correctly, can result in robust and interpretable decision-making solutions in any field

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